Dynamical Systems ~ Seminar Test

May 12, 2025

Name :

Group :

1. (5p) We consider the planar linear system $\dot{x} = 2x - 5y$, $\dot{y} = x - 2y$ and the function $H : \mathbb{R}^2 \to \mathbb{R}$, $H(x,y) = x^2 + 5y^2 - 4xy$.
   a) Find the solution of the system whose initial value is $\eta = (1,1)$.
   b) justify that H is a global first integral
   c) Specify the type and stability.

2. (3p) Find the solution of the IVP $tx' + 2x = t^2$, $x(1) = -2$.

3. (2p) Represent the phase portrait of $\dot{x} = 3x + x^2$. Specify
   Let $\varphi : \mathbb{R} \to \mathbb{R}$ be the solution whose initial value is $\eta = -1$.
   the values of $\lim_{t \to \infty} \varphi(t)$ and $\lim_{t \to -\infty} \varphi(t)$.

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1. (5p) We consider the linear planar system $\dot{x} = 2y$, $\dot{y} = -8x$.
   a) Justify that it has a global first integral (without finding it).
   b) Find a global first integral and represent its level curves.
   c) Represent the phase portrait of the system.
   d) Find the solution whose initial value is $\eta = (1, 1)$.

2. (3p) Find the solution of the IVP $tx' - 2x = \frac{1}{t}$, $x(1) = -\frac{1}{3}$.

3. (2p) Represent the phase portrait of $\dot{x} = 3x - x^2$. Let φ be the solution whose initial value is $\eta = -1$. If it is finite, specify the value of $\lim_{t \to \infty} \varphi(t)$ or $\lim_{t \to -\infty} \varphi(t)$.

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1. (5p) We consider the diff. eq $x'' + x = 2\cos t$.
a) Find a particular solution of the form $x_p(t) = t\,(a\cos t + b\sin t)$.
b) Let $\varphi : \mathbb{R} \to \mathbb{R}$ be the solution that verifies the initial conditions
$$x(0) = 0, \quad x'(0) = 0.$$
Find the expression of φ, represent its graph and describe its long-term behavior.
c) Represent in the cartesian plane xOy the curve of parametric equations $x = \varphi(t)$, $y = \varphi'(t)$, $t \in \mathbb{R}$.

2. (3p) Find the general solution of $x' + \frac{1}{t}x = \frac{1}{t}e^{-2t+1}$, $t > 0$.

3. (2p) If there is an attractor, find its basin of attraction, for $\dot{x} = -x(4 - x^2)$.

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1. (5p)